

PEE question set 3

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1 Question 1

Briefly explain the differences between Classification and Structured Output Prediction. Cite two application examples each paradigm

Classification can be understood as a problem where we have to label a certain sample or set of samples in one of the classes belonging to our class set. This set is finite and normally small, although nowadays the problem's size are growing and growing. The problem then can be seen as returning the label of the class that has the maximum probability given the input sample:

$$\arg \max_{c \in C} \Pr(c|x)$$

Where c and C are the predicted class and the full class set respectively and x is the input sample.

On the other hand, structured output prediction is a problem where the number of possible outputs is potentially infinite. It also has some kind of dependency in the output being for example a graph, formal grammar, sequence of chars or numbers with some order or exclusivity rule... Is this dependency what makes really the difference between this two approaches. Classification is agnostic of what has been predicted before, SOP must be aware in order to not break the rules or characteristics of the output.

Some examples of classification can be:

- Object recognition in images: such as MNIST, CIFAR, ImageNet... Where the system must tell whether the image contains the element represented by one label or the other.
- Music genre differentiation: where systems must learn the characteristics of each genre to put them apart and then classify the input samples correctly in Rock, Pop, Folk...

For the Structured Output Prediction, we can sample the following problems:

- Audio or image transcription: this includes given an image provide a description of what's in it or given some audio samples produce the corresponding subtitles.
- Text synthesis: where given a full document the system must produce a readable summary.

2 Question 2

Justify why the naive Bayes decomposition of Eq.(5) is adequate for karyotype recognition problem.

As seen in equation 5 of the theory dossier, the naive Bayes decomposition for the karyotype problem can be expressed as follows:

$$\Pr(x|h) = \Pr(x_1, ..., x_{22}|h_1, ..., h_{22}) \approx \prod_{i=1}^{22} \Pr(x_i|h_i)$$

Explained in words, we assume that given an hypothesis, the probability of the sample in a certain position, does only depend on itself, not on the past history. For the karyotype estimation, this approximation reduces the number of parameters to estimate, reducing also the computational cost of the algorithm.

Going deeper in the Karyotype example, this approximation makes particularly sense because the independence on the x exists, since the shape of the chromosome does not depend on the shape of the other chromosomes. The independence of h can be seen in a rather similar way. The representation (or label) given for a certain chromosome does not depend on the representation of the other. There's not a particular order that they must follow, so we can forget about the previous history when predicting the actual chromosome.

3 Question 3

Briefly explain all the steps and assumptions needed to derive Eq.(9) from Eq.(7)

In order to explain the derivation steps for this, we are going first to show the flow of equations and then explain them. This way the reader can go through the full equations without being interrupted by the text. Thus, the derivation goes as follows:

$$\hat{h} = \arg \max_{h \in H} \Pr(h|x, h', f) \quad (1)$$

$$\hat{h} = \arg \max_{h \in H} \Pr(h|x, h', f) = \arg \max_{h \in H} \frac{\Pr(h, x, h', f)}{\Pr(x, h', f)} \quad (2)$$

$$\hat{h} = \arg \max_{h \in H} \frac{\Pr(h, x, h', f)}{\Pr(x, h', f)} = \arg \max_{h \in H} \Pr(h, x, h', f) \quad (3)$$

$$\hat{h} = \arg \max_{h \in H} \Pr(h, x, h', f) = \arg \max_{h \in H} \Pr(d) \Pr(h'|d) \Pr(h|h', d) \Pr(x|h, h', d) \quad (4)$$

$$\hat{h} = \arg \max_{h \in H} \Pr(d) \Pr(h'|d) \Pr(h|h', d) \Pr(x|h, h', d) = \arg \max_{h \in H} \Pr(h|h', d) \Pr(x|h, h', d) \quad (5)$$

$$\hat{h} = \arg \max_{h \in H} \Pr(h|h', d) \Pr(x|h, h', d) = \arg \max_{h \in H} \Pr(h|h', d) \Pr(x|h) \quad (6)$$

As seen, we start from 1, if we apply the joint rule we get the result seen in 2. As we can see in 3, we are looking for the h that maximizes the probability, since the denominator does not depend on h , we can assume it's constant and remove it from the equation. Now we're doing a big assumption. We assume that f , the feedback from the outside, is deterministic. Doing so, we can define a function to decode this feedback: $d = d(f)$ and feed the problem with it. Replacing the feedback for its representation and applying the chain rule, we get what 4 displays. Again, in 5 we're maximizing over h , so $\Pr(d)$ and $\Pr(h'|d)$ are both constant so can be removed from the equation. That's what we've done at 5. Finally, since our actual hypothesis is based on the previous one and the decodification of the signal, we can assume that x can be considered independent from h' and d given a certain h . This way we get to our desired result: equation 6.

4 Question 6

Briefly explain all the steps and assumptions needed to derive Eq.(19) from Eq.(7)

As done in the previous exercise, we're going to provide first the derivation for the probability equations and then the explanation for each step.

$$\hat{h} = \arg \max_{h \in H} \Pr(h|x, h', f) \quad (7)$$

$$\hat{h} = \arg \max_{h \in H} \Pr(h|x, h', f) = \arg \max_{h \in H} \sum_d \Pr(h, d|x, h', f) \quad (8)$$

$$\hat{h} = \arg \max_{h \in H} \sum_d \Pr(h, d|x, h', f) = \arg \max_{h \in H} \sum_d \frac{\Pr(h, d, x, h', f)}{\Pr(x, h', f)} \quad (9)$$

$$\hat{h} = \arg \max_{h \in H} \sum_d \frac{\Pr(h, d, x, h', f)}{\Pr(x, h', f)} = \arg \max_{h \in H} \sum_d \Pr(h, d, x, h', f) \quad (10)$$

$$\hat{h} = \arg \max_{h \in H} \sum_d \frac{\Pr(h, d, x, h', f)}{\Pr(x, h', f)} \approx \arg \max_{h \in H} \max_{d \in D} \Pr(h, d, x, h', f) \quad (11)$$

$$\hat{h} \approx \arg \max_{h \in H} \max_{d \in D} \Pr(h, d, x, h', f) \approx \arg \max_{h \in H} \max_{d \in D} \Pr(h') \Pr(d|h') \Pr(f|h', d) \Pr(h|h', f, d) \Pr(x|h, h', f, d) \quad (12)$$

$$\hat{h} \approx (12) \approx \arg \max_{h \in H} \max_{d \in D} \Pr(d|h') \Pr(f|h', d) \Pr(h|h', d) \Pr(x|h) \quad (13)$$

We start from 7. In 8 we marginalize to make d appear so we can work with all the possible decodifications for the feedback signal. Then, in 9 we apply the joint rule. In 10, as the denominator is independent from the parameter for which we're maximizing, we can remove it from the equation. At 11 we have made an approximation. Instead of working with the sum for all possible decodifications, we take the mode as its value, this way we can ease the proceeding. At 12 we simply apply the chain rule. and finally to obtain 13 we have taken the following assumptions:

- $\Pr(h')$ is independent from both max values, so we can remove it from the equation since it is a constant value.
- The decodification already has information about the past history, since it depends on the user. Who has seen the past history. So we can remove h' from $\Pr(f|h', d)$
- Its obvious that the decodification comprehends the information about the feedback signal. So we can remove f from $\Pr(h|h', d, f)$
- Finally, our new hypothesis has been chosen based on the previous history, the signal and the decodification of the signal. Having this in mind, we can remove all d, f, h' from $\Pr(x|h, h', d, f)$

Note that we could have put both h and d in the $\arg \max$ operator and obtain the hypothesis and decodification that maximizes the probability. But to be consistent with the first equation, we have decided to obtain only $\hat{h}$.

5 Question 8

Briefly explain the concepts and main differences between Active and Passive interaction protocols

The difference between this two interaction protocols them relies on which hypothesis should be revisited by an expert. More deeply, in the **active** approach, the system has the capability to choose which action will be revisited. At each step, the system measures the level of confidence over all the elements. Then proposes for supervision the one with the lower confidence score. After the user reviews the classification and knowing if it was right or wrong, the system proceeds to label the rest of the samples based on the feedback and the past history. One of the main advantages of this approach is that the expert only has to revisit a small subset of the samples. On the other hand, this kind of systems are highly dependant on the ability of the system for rating and giving confidence scores.

In the **passive** approach is the user the one in charge of telling which samples should be revisited and which no. The revisit proceeding can be done in two different ways depending on the order:

- **Left-to-right:** This corresponds to that revisions that are performed starting for the top-left sample. The system assumes that all the samples placed at left of the revisited sample have been correctly predicted. Then, with the past history and the correction, the system predicts again the right part of the samples.
- **Desultory:** This corresponds to a kind of revision where there's no special order in which the expert corrects the system. This way the expert can correct the most flagrant error before the system even tries to correct other one.